

BRAC University
Department of Computer Science and Engineering

Examination: Quiz 04

Duration: 30 Minutes

Semester: Summer 26

Full Marks: 15

CSE420: Compiler Design

1. Consider the following SDT. Draw the parse tree, show the attribute calculations and generate the three address codes according to the SDT instructions.

Input String:

```
switch (a / b)
{
    case default : ab = b;
}
```

```
func genlabel()
{
    p = p + 1;
    return "L" || p ;
}
```

```
func label(l)
{
    return " _ " || l ;
}
```

SDT:

$P \rightarrow \mathbf{D\ S} \{ S.next = genlabel(); P.code = S.code \parallel label(S.next); \}$

$\mathbf{D} \rightarrow \{ p = 0; \}$

$S \rightarrow \mathbf{id = E} ; \{ S.code = E.code \parallel gen(top.get(id.lexeme) '=' E.addr); \}$

$E \rightarrow \mathbf{E_1 arith E_2} \{ E.addr = new Temp();$
 $E.code = E_1.code \parallel E_2.code \parallel gen(E.addr '=' E_1.addr arith.op E_2.addr); \}$

$E \rightarrow \mathbf{id} \{ E.addr = top.get(id.lexeme); E.code = ""; \}$

$S \rightarrow \mathbf{switch (E) \{ C \}} \{ C.next = S.next ; S.code = E.code \parallel gen(E.addr) \parallel C.cases \parallel C.default) \parallel C.code; \}$

$C \rightarrow \mathbf{case str : S_1 break ; C_1} \{ L = genlabel(); S_1.next = C.next; C_1.next = C.next;$
 $C.cases = str.lexeme \parallel " " \parallel L$
 $C.default = C_1.default;$
 $C.code = label(L) \parallel S_1.code \parallel gen('goto' C.next) \parallel C_1.code; \}$

$C \rightarrow \mathbf{case default : S_2} \{ L = genlabel(); S_2.next = C.next; C.default = L$
 $C.code = label(L) \parallel S_2.code; \}$

$C \rightarrow \epsilon \{ C.default = C.next; C.code = ""; \}$